

Education

Master of Science in Computer Science

Aug '21 - May '23 (expected)

Georgia Institute of Technology, Atlanta, Georgia

Bachelor of Technology (Honours) in Computer Science and Engineering

Jul '13 - May '17

International Institute of Information Technology, Hyderabad, India

- GPA: **9.59** / 10, Class Rank: **2** / 275
- **Best All-rounder Gold Medalist** for demonstrating excellence in academics and extra-curricular activities.

Work Experience

Staff Software Engineer, Indeed.com

Apr '21 - Jul '21

- Worked on **Indeed Apply** - a platform that creates and delivers **millions** of job applications to employers worldwide everyday.
- **Team lead** for a 6-member engineering sub-team, planned the team's work and drove the deliverables via other engineers with mentorship.
- **Technical lead** for a 30-member engineering team working on end-to-end experience of the job application system involving multiple frontend, API and backend components.
- **Founder and lead, GraphQL Guild** - Led a group of 20 engineers from different teams, collected and maintained knowledge about GraphQL and recommended best practices to teams across Indeed.

Senior Software Engineer, Indeed.com

Oct '19 - Mar '21

- **Designed and developed an API** using GraphQL that evaluates the quality of job applications to inform applicants about a possible mismatch with the job, reducing **10 million** low quality applications every month.
- **Led a multi-quarter project** to redesign the Indeed Apply frontend that improved hires by around 4%.
- **Mentored** fellow engineers on frameworks like React.js, Spring and Node.js.

Software Engineer, Indeed.com

Jul '17 - Sep '19

- **Collaborated with Glassdoor**, played a key role in enabling Indeed Apply for jobs on Glassdoor.
- Improved release velocity and maintainability by breaking down the web application into **reliable micro- frontend services** serving UI components.
- Worked on **Indeed Placement** - a platform to easily hire fresh college graduates in India.
 - **Selected for funding** of \$500K to work with the internal product accelerator.
 - Conducted market research by visiting employers and colleges for gathering and prioritizing requirements.
 - Handled complete product development single-handedly, and provided technical direction to the team.

Patent

Multiple Pane Web Display with Dynamic Content

Filed on December 11, 2019

Improve job application experience for job seekers on Indeed.com by displaying job description and application form on a single page with an inset online resume editor. (*U.S. Patent Application No.: 16/711,299*)

Technical Skills

Programming Languages: Java, Python, Javascript, C++ (with STL), C**Frameworks, Databases:** Spring, React.js, GraphQL, Node.js, Django, MongoDB, MySQL**Tools:** MATLAB, Robot Operating System, Git

Research Experience

Graduate Researcher, Georgia Tech

Aug '21 - Present

Advisor: Prof. Harish Ravichandar

- Learning dexterous manipulation skills from demonstrations to improve sample efficiency, explainability and theoretical guarantees over the existing state-of-the-art algorithms.

Visiting Scholar, Robotics Embedded Systems Lab, University of Southern California

May '16 - Jul '16

Advisors: Prof. Gaurav S. Sukhatme, Dr. Oliver Kroemer

- Exploring CNN based Feature Transfer for Robot Affordances [[poster](#)]
- Used *TensorFlow* to learn CNN on previously annotated affordance data, and Transfer Learning to learn new affordances (with few samples).
- Annotated *grasp* and *push* data by manipulating PR2 using *ROS*.

Advisor: *Prof. K. Madhava Krishna*

- Incremental Surface Reconstruction and Tracking: Dense CRF for semantic motion labelling, Truncated Signed Distance Field, Ray casting and Medioni's transform for 3D reconstruction of an outdoor scene, using *Kinect Fusion: InfiniTAM* library. (C++)
- Driverless Car Challenge: Part of development of driverless car, worked on camera calibration, motion segmentation, application of B-Spline curve fitting, and obstacle detection.
- Learning affordances from robot environment: Created an environment in simulation for robots to interact with and learn affordances using Reinforcement Learning.

Teaching Experience

Teaching Assistant at IIIT Hyderabad

- **Computer Vision** with Prof. Anoop M. Namboodiri *Jan '17 - May '17*
- **Computer Programming** with Prof. Anoop M. Namboodiri *Aug '16 - Dec '16*
- **Computer Systems Organization** with Prof. Govindarajulu Regeti *Jan '16 - May '16*
- **Digital Logic and Processors** with Prof. K. Madhava Krishna, Dr. V. Eathakota *Aug '15 - Dec '15*

Achievements

- Awarded **Dean's Merit List** in all semesters for being among the **top 5%** students.
- Selected among the **top 20** students from **all over India** for the Viterbi-India program by The Indo-US Science and Technology Forum.
- Ranked **51st** from over 1500 teams in **ACM ICPC Asia-Amritapuri 2015-16 Regionals**.
- Qualified for ACM ICPC Regionals for 3 years in a row, securing **top 100** online and onsite ranks.
- Secured **All India Rank 309** in Joint Entrance Examination (Mains) 2013 among 1.4 million candidates.
- Among the **top 0.1%** students who received a **merit certificate** in Computer Science awarded by the Central Board of Secondary Education in Class 12.

Projects

Analyzing Facial Features for Recommendation, Statistical Methods in AI *Sep '15 - Nov '15*

- Learn user preferences for various facial images, suggest faces based on learning, use of Dense-SIFT, PCA, HOG, GIST algorithms and SVM classifier. [[report](#)] (team size: 2)

Obstacle Avoidance and Path Planning, Mobile Robotics *Sep '15*

- Using Bernstein curve fitting, used the concept of configuration space, visibility graph; implemented on TurtleBot. (MATLAB)

Ultimate Tic-Tac-Toe Bot, Artificial Intelligence *Mar '15*

- Implemented bot for playing UT3, used Alpha-Beta pruning, hashing, IDFS, finished 3rd among 100 teams. [[github](#)] (team size: 2)

Robot Central Management Server, Structured System Analysis and Design *Aug '14 - Nov '14*

- Using Django, portal to control robots in a workspace, commands executed using ROS at the backend, basic path planning algorithm for Tech Mahindra. (team size: 4)

Relevant Courses

Systems: Distributed Systems, Database Systems, Advanced Algorithms, Advanced Computer Networks

Mathematics: Discrete Mathematics, Functional Analysis, Differential Equations, Optimization Methods

Fundamentals: Data Structures, Algorithms, Computer Systems Organization, Operating Systems

Machine Learning: Machine Learning, Computer Vision, Statistical Analysis in AI

Others: Computer Graphics, Mobile Robotics, Artificial Intelligence, Structured System Analysis and Design

Extra-curricular activities

Students' Parliament, 2015-17: Elected **Speaker** of Parliament meetings, moderated discussions on matters of student life on campus, and allocated tasks among the members. Served as liaison between faculty, administration and the student community, especially in cases of *disciplinary concerns*.

Overall Coordinator, Felicity Threads, 2016: Led a team of around 40 people to organize college's technical fest including international contests for competitive programming, mathematical quizzing, ML, AI etc. having over 10K participants globally.

Member, Student Induction Body, 2015-17: Helped freshman students in the admission process by answering their queries in liaison with the admissions office, arranged for Welcome Kit, and Orientation tours.

Cultural Representative, Agni House, 2014-15: Led dance troupes, directed and participated in skits, for cultural performances in Inter-house competitions.

Teaching Volunteer at Ashakiran, 2015-16: Tutored secondary school kids from nearby slums.